



DEPARTMENT OF ELECTRICAL AND COMPUTER ENGINEERING

AUTOMATIC DOOR SYSTEM USING DISCRETE H-BRIDGE DRIVER

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1. LITERATURE REVIEW

This project integrates concepts from analog electronics, digital control, DC motor drives, and ultrasonic sensing to implement a low-cost automatic door system using an Arduino Uno and a discrete H-bridge. DC motors are widely used in automation because of their simple speed and direction control by means of voltage polarity and pulse-width modulation (PWM). Standard electronics textbooks describe the use of bipolar junction transistors (BJTs) as power switches capable of driving low-to-medium-power DC motors when properly biased into saturation.

The H-bridge is a classic topology that allows bidirectional current flow through a DC motor by appropriately switching four power devices arranged as a bridge. Traditional implementations use BJTs or MOSFETs as the active elements, combined with base- or gate-drive stages that ensure saturation and protect the microcontroller from large currents. In the educational context, discrete H-bridge designs are useful to visualize how individual transistors interact to reverse the polarity applied to the motor.

Modern motor-driver ICs, such as the L298N and L293D, integrate protection diodes, current-limiting and thermal protection, simplifying the design of bidirectional DC motor controllers at the expense of additional voltage drop and efficiency loss. Likewise, MOSFET-based H-bridges are preferred in many industrial designs because of their low on-resistance and reduced conduction losses compared with BJTs. In this project we deliberately selected discrete TIP31C (NPN) and TIP32C (PNP) transistors to emphasize the underlying power-electronics concepts and to compare their performance and limitations against these more advanced alternatives.

Ultrasonic distance sensors such as the HC-SR04 are commonly used for non-contact detection in robotics and access-control applications. They estimate distance by measuring the round-trip time of an ultrasonic pulse and are inexpensive, easy to interface with microcontrollers, and sufficiently accurate for short-range detection. However, their readings can be affected by irregular surfaces, incidence angles, and environmental noise. These characteristics make them suitable for a didactic automatic door, while also providing an opportunity to analyze measurement noise and system limitations.

2. INTRODUCTION

2.1 Problem Statement

Many access-control systems require the automatic opening and closing of doors when a person or object approaches the entrance. Commercial solutions typically rely on

integrated motor drivers and proprietary controllers, which limits their use as educational platforms. For undergraduate students, it is desirable to develop a low-cost prototype that clearly exposes the interaction between sensing, control logic, and discrete power-electronics hardware.

The specific problem addressed in this project is the design and implementation of an automatic door system capable of opening and closing a small door when an object is detected within a predefined distance. The system must be implemented using an Arduino Uno, an HC-SR04 ultrasonic sensor, and a discrete H-bridge constructed from TIP31C and TIP32C transistors, while operating from a simple 9 V battery supply for the motor stage.

2.2 Objectives

2.2.1 General Objective

To design, build, and experimentally validate an automatic door system that uses an Arduino-based control unit, an ultrasonic distance sensor, and a discrete H-bridge driver to perform reliable bidirectional control of a DC motor for opening and closing a door.

2.2.2 Specific Objectives

- To implement an H-bridge using TIP31C (NPN) and TIP32C (PNP) transistors capable of driving a 9 V DC motor in both directions.
- To interface an HC-SR04 ultrasonic sensor with the Arduino Uno and configure it to detect objects within a predefined distance threshold.
- To develop a finite state machine in the Arduino that controls the door states (CLOSED, OPENING, OPENED, CLOSING) based on the measured distance and time intervals.
- To estimate and experimentally verify the activation time required for the motor to rotate approximately 90° for opening and closing the door.
- To characterize electrical and mechanical performance, including transistor voltage drops, base currents, and motor current consumption, and to identify the main limitations of the prototype.

2.3 Constraints

- First, the power stage was required to use discrete BJTs rather than integrated motor-driver ICs, which limits efficiency and increases voltage drops across the H-bridge.
- Second, the motor supply had to be provided by a single 9 V battery, introducing significant internal resistance and voltage sag under load.
- Third, the mechanical structure was implemented using a cardboard model house, which introduces friction, misalignment, and structural deformation.
- Finally, the system was constrained by the capabilities of the Arduino Uno, including limited processing resources and the need to share 5 V logic levels with the sensor and BJT base-drive network.

5. DESIGN DETAILS

3.1 Subsystems

The proposed system can be divided into four main subsystems: (1) the sensing subsystem, based on the HC-SR04 ultrasonic sensor used to detect the presence of an object near the door; (2) the control subsystem, implemented with an Arduino Uno R3 that executes the finite state machine and generates digital control signals; (3) the power-driver subsystem, formed by a discrete H-bridge using TIP31C and TIP32C transistors that deliver bidirectional current to the DC motor; and (4) the mechanical subsystem, which consists of a 9 V DC motor mechanically coupled to a small cardboard door mounted on a model house. These subsystems interact to convert distance measurements into controlled door motion.

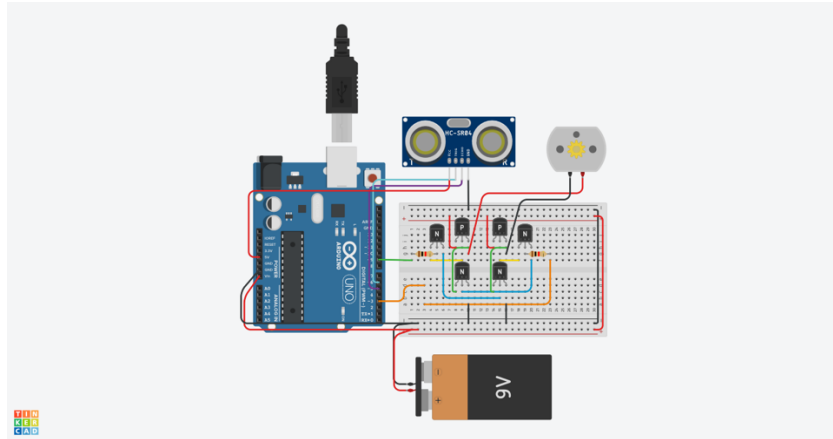


Figure 1. TinkerCAD simulation of the complete automatic door system, including the Arduino Uno, HC-SR04 sensor, and H-bridge driver.

3.2 Power Supply

The prototype uses two separate power domains. The Arduino Uno R3 and the HC-SR04 sensor are powered from the 5 V USB supply, which provides a stable voltage for logic and timing operations. The motor and power-driver stage are supplied from an independent 9 V battery connected to the H-bridge. A common ground reference is shared between both domains to ensure correct signal levels at the transistor bases. The use of a 9 V alkaline battery simplifies the design but introduces limitations related to its relatively high internal resistance and limited current capability. Under motor load, the effective voltage available to the H-bridge decreases, especially when combined with the voltage drops across the TIP31C and TIP32C transistors in saturation.

3.3 Driver Design (H-Bridge)

The H-bridge was implemented using two TIP32C PNP transistors as the high-side devices and four TIP31C NPN transistors on the low-side branches. Each BJT base is connected through a 1 k Ω resistor to limit base current and protect the Arduino pins.

The DC motor is connected between the two central nodes of the bridge, so that activating opposite pairs of transistors reverses the polarity applied to the motor and hence its direction of rotation.

On each side of the bridge, an NPN transistor driven directly by the Arduino acts as an inverting driver that turns the corresponding PNP high-side device on or off. When the “forward” signal is asserted, one diagonal pair of transistors saturates, connecting the motor terminals to +9 V and ground in the forward orientation. When the “reverse” signal is asserted, the opposite diagonal pair is activated, inverting the motor polarity and causing the door to close. This configuration provides a clear demonstration of how BJTs can be combined to implement a discrete motor-driver stage.

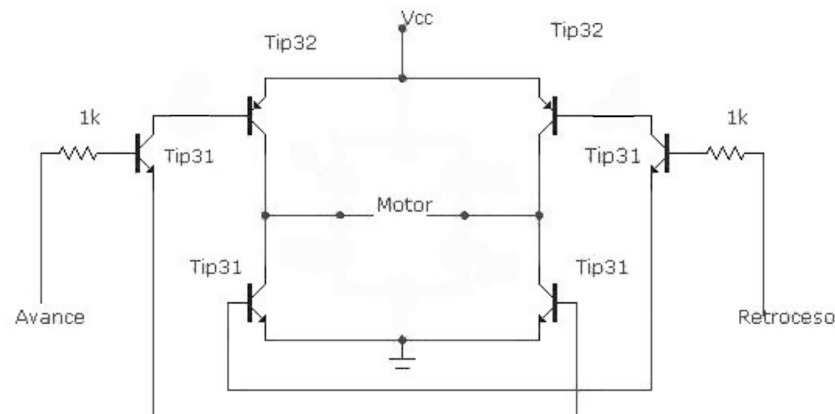


Figure 2. H-bridge used in the project implemented with discrete TIP31C/TIP32C transistors.

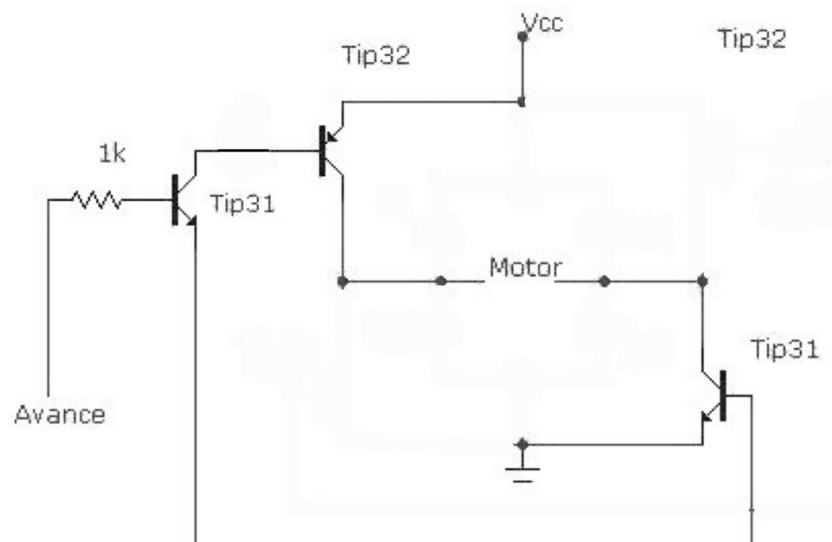


Figure 3. Left branch of the H-bridge for the forward mode. The TIP31 transistor with a 1 k Ω base resistor acts as an inverting driver for the upper TIP32 transistor, which connects the motor to VCC.

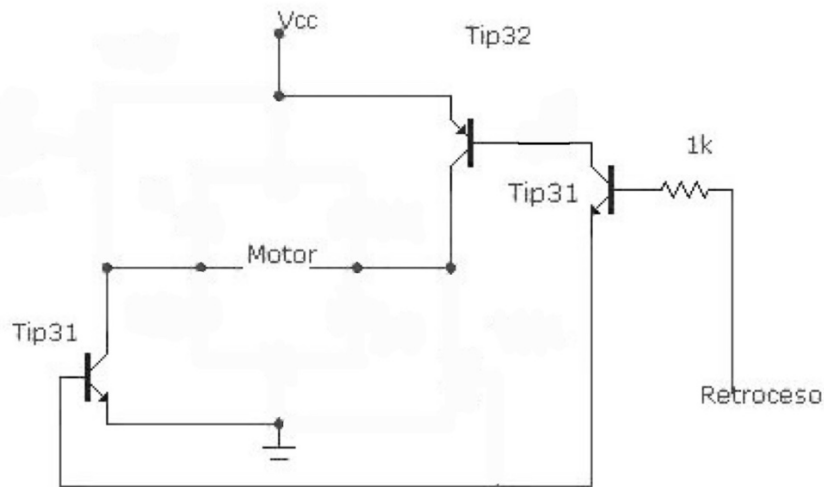


Figure 4. Right branch of the H-bridge for the reverse mode. In this case, the TIP31 transistor with a 1 k Ω base resistor drives the opposite TIP32 device, inverting the polarity applied to the motor to close the door.

3.4 Software Design

The control algorithm is implemented on the Arduino Uno as a finite state machine with four states: CLOSED, OPENING, OPENED, and CLOSING. In the CLOSED state, the motor is off and the door remains fully closed while the system continuously reads the distance from the HC-SR04 sensor. When the measured distance falls below a predefined threshold, the controller transitions to the OPENING state.

In the OPENING state, the Arduino activates the H-bridge in the forward direction, allowing the motor to rotate for approximately 400 ms, which corresponds to a 90° rotation under the chosen PWM conditions. After this interval, the system enters the OPENED state, where the motor is turned off and a timer maintains the door open for a specified waiting time. Once this timer expires, the controller transitions to the CLOSING state, in which the H-bridge reverses the motor polarity for another 400 ms to return the door to the closed position. The system then returns to the CLOSED state and waits for the next object detection.

Code:

```
// Pines
const int TRIG_PIN = 6;      // ultrasónico TRIG (PWM-capable también, pero lo usamos
                             // digital normal)
const int ECHO_PIN = 5;     // ultrasónico ECHO
const int IN1 = 3;          // <-- debe ser pin PWM!
const int IN2 = 9;          // <-- debe ser pin PWM!

// Parámetros
const long DIST_THRESHOLD_CM = 20;
const unsigned long AUTO_CLOSE_DELAY = 3000;
```

```

const unsigned long DEAD_TIME_MS = 500;
const unsigned long MOVE_TIMEOUT_MS = 400;

// Estados
enum State {CLOSED, OPENING, OPENED, CLOSING};
State state = CLOSED;

// Tiempos
unsigned long stateStartTime = 0 ;
unsigned long lastPresenceTime = 0;

// Velocidades (0-255)
const int SPEED_OPEN  = 40; // qué tan rápido abre
const int SPEED_CLOSE = 40; // un poco más lento cerrando por seguridad

void setup() {
    pinMode(TRIG_PIN, OUTPUT);
    pinMode(ECHO_PIN, INPUT);

    pinMode(IN1, OUTPUT);
    pinMode(IN2, OUTPUT);

    stopMotor();
    Serial.begin(9600);

    state = CLOSED;
    stateStartTime = millis();
}

void loop() {
    unsigned long now = millis();
    long dist = readUltrasonicCM();

    Serial.print("Dist(cm): ");
    Serial.print(dist);
    Serial.print("  State: ");
    Serial.println(state);

    switch (state) {

        case CLOSED:
            if (dist < DIST_THRESHOLD_CM) {
                openDoor();
            }
            break;

        case OPENING:
            if (now - stateStartTime >= MOVE_TIMEOUT_MS) {

```

```

        stopMotor();
        state = OPENED;
        stateStartTime = now;
        lastPresenceTime = now;
    }
    break;

case OPENED:
    if (dist < DIST_THRESHOLD_CM) {
        lastPresenceTime = now; // alguien todavía ahí
    }
    if ((now - lastPresenceTime) >= AUTO_CLOSE_DELAY) {
        closeDoor();
    }
    break;

case CLOSING:
    // safety: si alguien se mete, reabrimos
    if (dist < DIST_THRESHOLD_CM) {
        openDoor();
        break;
    }
    if (now - stateStartTime >= MOVE_TIMEOUT_MS) {
        stopMotor();
        state = CLOSED;
        stateStartTime = now;
    }
    break;
}

delay(50);
}

// ----- Funciones de movimiento -----

void stopMotor() {
    // apagar ambas ramas -> motor libre/parado
    analogWrite(IN1, 0);
    analogWrite(IN2, 0);
}

// gira en la dirección "abrir"
void driveOpen(int speedVal) {
    // Sentido abrir = IN1 activo en PWM, IN2 apagado
    analogWrite(IN1, speedVal);
    analogWrite(IN2, 0);
}

```

```

// gira en la dirección "cerrar"
void driveClose(int speedVal) {
    // Sentido cerrar = IN2 activo en PWM, IN1 apagado
    analogWrite(IN1, 0);
    analogWrite(IN2, speedVal);
}

void openDoor() {
    stopMotor();
    delay(DEAD_TIME_MS);

    driveOpen(SPEED_OPEN);

    state = OPENING;
    stateStartTime = millis();
}

void closeDoor() {
    stopMotor();
    delay(DEAD_TIME_MS);

    driveClose(SPEED_CLOSE);

    state = CLOSING;
    stateStartTime = millis();
}

// ----- Sensor ultrasónico -----
long readUltrasonicCM() {
    digitalWrite(TRIG_PIN, LOW);
    delayMicroseconds(2);
    digitalWrite(TRIG_PIN, HIGH);
    delayMicroseconds(10);
    digitalWrite(TRIG_PIN, LOW);

    long duration = pulseIn(ECHO_PIN, HIGH, 30000UL);
    if (duration == 0) return 999;
    return duration / 58L;
}

```

4. DESIGN VERIFICATION

4.1 Testing

The system was tested experimentally to verify correct operation of the sensing, control, and power-driver subsystems. First, the HC-SR04 sensor was characterized by

measuring the reported distance to a flat target at different positions, confirming reliable detection within the range required to trigger the door. Second, the timing of the OPENING and CLOSING states was adjusted so that a 400 ms activation produced an approximate 90° rotation of the door, based on the measured motor speed under the selected PWM duty cycle.

Electrical measurements focused on the H-bridge performance. The voltage drop across individual TIP31C and TIP32C transistors in saturation was measured to be approximately 0.8 V in each device, resulting in a total drop of about 1.6 V across the conduction path of the H-bridge. Base currents around 4.3 mA were obtained using 1 k Ω base resistors and a 5 V logic level, which is sufficient to saturate the transistors for the low-power DC motor used in the prototype.

Motor current was also monitored during operation. The no-load current was in the range of 600–700 mA, while the typical current during door opening remained between 750–800 mA, which is within the safe operating region of the selected BJTs but represents a significant load for a 9 V alkaline battery.

4.2 Evaluation of Results

The experimental results confirm that the system can reliably detect an approaching object and automatically open and close the door using the discrete H-bridge. The measured activation time of 400 ms closely matches the theoretical time required to achieve a 90° rotation based on the effective motor speed, validating the proposed timing model. The finite state machine correctly enforces the CLOSED, OPENING, OPENED, and CLOSING states without unintended transitions.

Nonetheless, several limitations were identified. The voltage drops across the BJTs reduce the effective voltage applied to the motor, especially when the 9 V battery voltage decreases under load. The transistors exhibit noticeable heating during prolonged operation due to conduction losses, and the mechanical structure of the cardboard model introduces friction and alignment issues that affect motion repeatability. Additionally, the ultrasonic sensor readings show fluctuations of ± 1 –3 cm under certain conditions, such as irregular surfaces or strong environmental vibrations. These observations highlight opportunities for improvement in future versions of the design.

5. COST ANALYSIS

The prototype was designed to be a low-cost educational project using widely available components.

Table 1 summarizes the main items used in the implementation, including the Arduino Uno board, the HC-SR04 ultrasonic sensor, the 9 V DC motor, the discrete TIP31C/TIP32C transistors, and the mechanical materials required to build the model house and door.

As shown in Table 1, the total cost of the prototype is approximately 60 USD.

This value is within the typical budget allocated for undergraduate design projects and demonstrates that an automatic door system with sensing, control, and discrete power electronics can be implemented with relatively low investment.

The low overall cost makes the design suitable for replication in future ELEN-330 laboratory sections or as a demonstrator for basic automation concepts.

Table 1 – Bill of Materials

Item	Qty	Unit Price (USD)	Total (USD)
Arduino Uno (or compatible board)	1	20.00	20.00
9V DC motor	1	10.00	10.00
HC-SR04 ultrasonic sensor	1	4.00	4.00
TIP31C / TIP32C BJTs	6	0	0
Protoboard shield	1	8.00	8.00
Resistors, wires, connectors	1	5.00	5.00
Mechanical materials (cardboard, glue, screws, etc.)	1	15.00	15.00
		Total	62.00

6. CONCLUSIONS AND FUTURE WORK

6.1 Conclusions

As a group, we successfully designed and implemented an automatic door system using an Arduino Uno, an HC-SR04 ultrasonic sensor, and a discrete H-bridge built from TIP31C/TIP32C transistors. The system demonstrated stable operation and reliable detection of approaching objects, automatically opening and closing the door with an approximate 90° rotation. The project allowed us to integrate concepts from analog electronics, digital control, sensors, and power-electronics design in a tangible prototype.

Experimental measurements showed that the timing model used to select a 400 ms activation interval for the motor is consistent with the measured motor speed, confirming the validity of the theoretical analysis. At the same time, we observed the impact of BJT saturation voltage on the available motor voltage and efficiency, as well as the thermal limitations of discrete transistors when driving relatively high currents from a 9 V battery.

Overall, the prototype meets its educational objective by clearly illustrating the interaction between sensing, control algorithms, and discrete power-driver hardware in a low-cost automatic door system.

6.2 Future Work

Future improvements to the system include replacing the discrete BJT H-bridge with a MOSFET-based driver or a dedicated motor-driver IC, in order to reduce conduction losses, improve efficiency, and increase motor torque. The power supply could be upgraded from a 9 V alkaline battery to a Li-ion cell with a DC-DC converter to provide a more stable voltage under load.

On the sensing and control side, additional filtering and signal-processing techniques could be implemented to mitigate noise in the ultrasonic distance measurements. Mechanically, a more rigid structure and precise hinges would reduce friction and improve repeatability. Finally, a custom PCB could be designed to integrate the H-bridge, sensor interface, and Arduino connections into a more compact and robust module suitable for long-term demonstrations and further extensions of the project.

7. REFERENCES

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